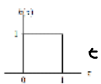


Convolution

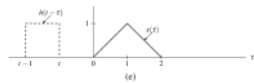
Saturday, September 28, 2024 8:30 AM

Example 4.3. Compute the quantity $x * h$, where

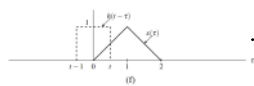
$$x(t) = \begin{cases} 0 & t < 0 \\ t & 0 \leq t < 1 \\ -t+2 & 1 \leq t < 2 \\ 0 & t \geq 2 \end{cases} \quad \text{and} \quad h(t) = u(t) - u(t-1).$$



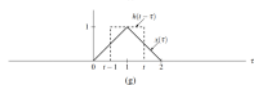
$$x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau) h(t-\tau) d\tau$$



for $t < 0$ $x(t) * h(t) = 0$



for $0 < t < 1$ $x(t) * h(t) = \int_0^t (\tau)(1) d\tau = \frac{1}{2} t^2$



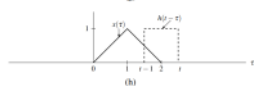
for $1 \leq t < 2$

$$x * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{t-1}^t \tau d\tau + \int_1^t (-\tau+2) d\tau$$

$$= \left[\frac{1}{2} \tau^2 \right]_{t-1}^t + \left[-\frac{1}{2} \tau^2 + 2\tau \right]_1^t$$

$$= \frac{1}{2} - \frac{1}{2} (t-1)^2 - \frac{1}{2} t^2 + 2t - \left[-\frac{1}{2} + 2 \right]$$

$$= -t^2 + 3t - \frac{3}{2}$$



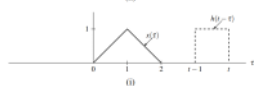
for $2 \leq t < 3$

$$x * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{t-1}^2 (-\tau+2) d\tau$$

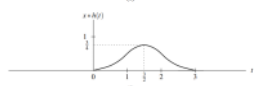
$$= \left[-\frac{1}{2} \tau^2 + 2\tau \right]_{t-1}^2$$

$$= 2 - \left[-\frac{1}{2} t^2 + 2t - \frac{3}{2} \right]$$

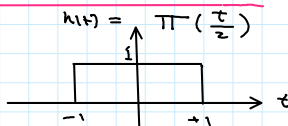
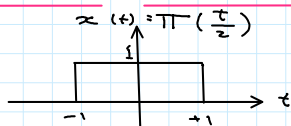
$$= \frac{1}{2} t^2 - 3t + \frac{7}{2}$$



for $t \geq 3$ is 0

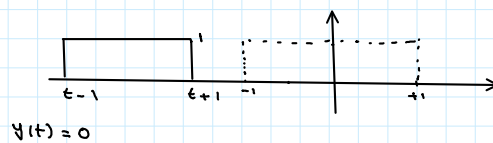


$$x * h(t) = \begin{cases} 0 & t < 0 \\ \frac{1}{2} t^2 & 0 \leq t < 1 \\ -t^2 + 3t - \frac{3}{2} & 1 \leq t < 2 \\ \frac{1}{2} t^2 - 3t + \frac{7}{2} & 2 \leq t < 3 \\ 0 & t \geq 3 \end{cases}$$

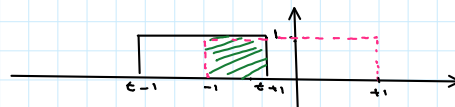


$$\int_{-\infty}^{+\infty} x(\tau) h(t-\tau) d\tau$$

I) $t+1 < -1 \Rightarrow t < -2$



II) $t+1 \geq 1$ & $t-1 \leq -1$
or $-2 \leq t < 0$



$$y(t) = \int_{-1}^{t+1} (1)(1) d\tau = t+2$$

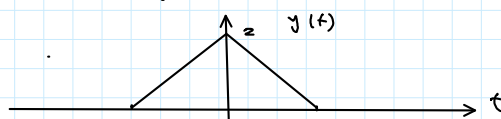
III) $t+1 > 1$ & $t-1 \leq +1$

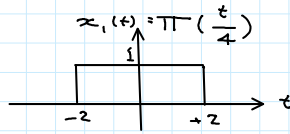
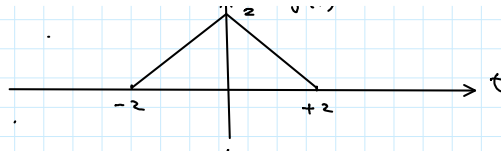


$$0 \leq t \leq 2$$

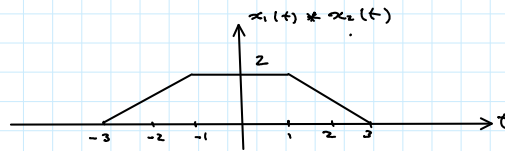
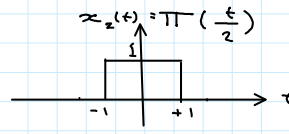
$$y(t) = \int_{t-1}^{+1} (1)(1) d\tau = 2-t$$

IV) $t-1 > +1 \Rightarrow t > 2$ $y(t) = 0$

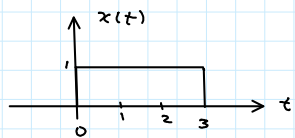




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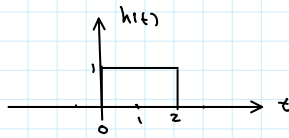


Evaluate $y(t) = x(t) * h(t)$, shown below, by analytical technique (do not use graphical method)



$$x(t) = u(t) - u(t-3)$$

*



$$h(t) = u(t) - u(t-2)$$

$$\begin{aligned} y(t) &= x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau) h(t-\tau) d\tau \\ &= \int_{-\infty}^{+\infty} \underbrace{[u(\tau) - u(\tau-3)]}_{x(\tau)} \underbrace{[u(t-\tau) - u(t-\tau-2)]}_{h(t-\tau)} d\tau \\ &= \int_{-\infty}^{+\infty} \underbrace{u(\tau) \cdot u(t-\tau)}_{\text{I}} d\tau - \int_{-\infty}^{+\infty} \underbrace{u(\tau) \cdot u(t-2-\tau)}_{\text{II}} d\tau \\ &\quad - \int_{-\infty}^{+\infty} \underbrace{u(\tau-3) \cdot u(t-\tau)}_{\text{III}} d\tau + \int_{-\infty}^{+\infty} \underbrace{u(\tau-3) \cdot u(t-2-\tau)}_{\text{IV}} d\tau \end{aligned}$$

Since:

$$\text{I} \quad u(\tau) u(t-\tau) = \begin{cases} 1 & 0 < \tau < t \text{ \& } t > 0 \\ 0 & \text{o/w} \end{cases} \quad \text{(cloud: } u(\tau) \text{)}$$

$$\text{II} \quad u(\tau) u(t-2-\tau) = \begin{cases} 1 & 0 < \tau < t-2 \text{ \& } t > 2 \\ 0 & \text{o/w} \end{cases} \quad \text{(cloud: } u(\tau-2) \text{)}$$

$$\text{III} \quad u(\tau-3) u(t-\tau) = \begin{cases} 1 & 3 < \tau < t \text{ \& } t > 3 \\ 0 & \text{o/w} \end{cases} \quad \text{(cloud: } u(\tau-3) \text{)}$$

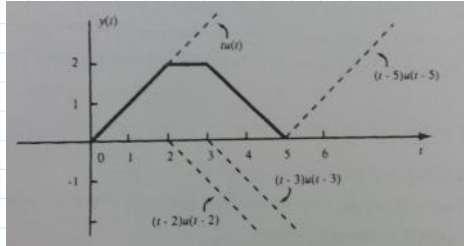
$$\text{IV} \quad u(\tau-3) u(t-2-\tau) = \begin{cases} 1 & 3 < \tau < t-2 \text{ \& } t > 5 \\ 0 & \text{o/w} \end{cases} \quad \text{(cloud: } u(\tau-5) \text{)}$$

$$\text{I} \quad u(\tau-3) = \begin{cases} 1 & \tau > 3 \\ 0 & \tau < 3 \end{cases}$$

$$\text{II} \quad u(t-2-\tau) = \begin{cases} 1 & t-2-\tau > 0 \Rightarrow \tau < t-2 \\ 0 & \text{o/w} \end{cases}$$

II & I for $t > 5$

$$\begin{aligned}
 y(t) &= \left(\int_0^t d\tau \right) u(t) - \left(\int_0^{t-2} d\tau \right) u(t-2) \\
 &\quad - \left(\int_3^t d\tau \right) u(t-3) + \left(\int_3^{t-2} d\tau \right) u(t-5) \\
 &= t u(t) - (t-2) u(t-2) - (t-3) u(t-3) \\
 &\quad + (t-5) u(t-5)
 \end{aligned}$$



Fourier Series Alternative:

$f(t)$ is periodic $T = \frac{2\pi}{\omega_0}$

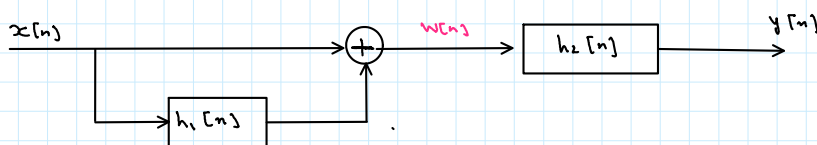
$$f(t) = \sum_{n=-\infty}^{+\infty} C_n e^{jk\omega_0 t}$$

or $f(t) = d_0 + \sum_{n=1}^{+\infty} \left(a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) \right)$

$$C_n = \begin{cases} \frac{1}{2} (a_n - j b_n) & n = 1, 2, 3, \dots \\ d_0 & n = 0 \\ \frac{1}{2} (a_{-n} + j b_{-n}) & n = -1, -2, -3, \dots \end{cases}$$

$$\begin{aligned}
 a_n &= 2 \operatorname{Re} \{ C_n \} & |C_n| &= \sqrt{a_n^2 + b_n^2} \\
 b_n &= -2 \operatorname{Im} \{ C_n \} & \theta_n &= -\arctan \left(\frac{b_n}{a_n} \right)
 \end{aligned}$$

Properties of Convolution:



$$h_1[n] = \beta \delta[n-1] \quad h_2[n] = \alpha^n u[n]$$

$$\begin{aligned}
 w[n] &= x[n] + (x[n] * h_1[n]) & x[n] &\rightarrow h_{\text{overall}}[n] \rightarrow y[n] \\
 y[n] &= w[n] * h_2[n] \\
 &= (x[n] + (x[n] * h_1[n])) * h_2[n] & y[n] &= h_{\text{overall}}[n] * x[n] \\
 &= x[n] * h_2[n] + (x[n] * h_1[n]) * h_2[n] \\
 &= x[n] * (h_2[n] + h_1[n] * h_2[n])
 \end{aligned}$$

overall $h[n]$

$$h_{\text{overall}}[n] = h_2[n] + h_1[n] * h_2[n] \\ = \alpha^n u[n-1] + \beta \alpha^{n-1} u[n-1]$$

Procedure II:

$$w[n] = x[n] + x[n] * h_1[n]$$

$$\text{Let } x[n] = \delta[n]$$

$$w[n] = \delta[n] + \delta[n] * h_1[n]$$

$$h_2[n] = w[n] * h_2[n]$$

$$= (\delta[n] + \underbrace{\delta[n] * h_1[n]}_{h_1[n]}) * h_2[n]$$

$$= h_2[n] + h_1[n] * h_2[n]$$

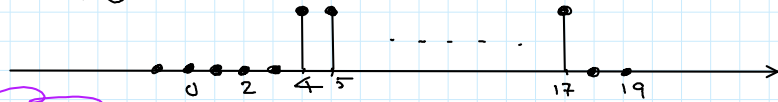
More example on Convolution:

LTI sys. input:

$$x[n] = u[n-3]$$

impulse response: $h[n] = (-\frac{1}{3})^n (u[n-4] - u[n-18])$

what is $y[n]$ at $n=8$



Note:

$$h[n] * u[n] = \sum_{k=-\infty}^n h[k]$$

$$h[n] * u[n-n_0] = \sum_{k=-\infty}^{n-n_0} h[k]$$

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} h[k] x[n-k]$$

$$y[n] = u[n-n_0] * h[n] = \sum_{k=-\infty}^{n-n_0} h[k]$$

$n_0 = 3$

$$\therefore y[8] = \sum_{k=-\infty}^{8-3} h[k] = \sum_{k=-\infty}^5 h[k]$$

بالنسبة لـ $y[8]$

$$\left(h[n] = (-\frac{1}{3})^n (u[n-4] - u[n-18]) \right)$$

فقط $h[n]$ من $n=4$ إلى $n=17$

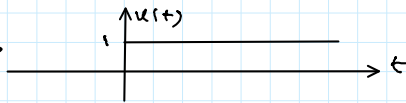
$$\text{جواب: } y[8] = \sum_{k=4}^5 h[k] = h[4] + h[5] = (-\frac{1}{3})^4 + (-\frac{1}{3})^5 = 2(\frac{1}{3})^5$$

LTI Systems: Finding impulse Response:

$$h(t) = y(t) \Big|_{\text{when } x(t) = \delta(t)}$$

if input-output relationship:

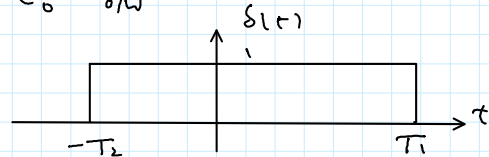
$$I) \quad y(t) = \int_{-\infty}^t x(\tau) d\tau \quad \text{LTI} \quad x(\tau) = \delta(\tau)$$

$$\Rightarrow h(t) = u(t) \Rightarrow$$


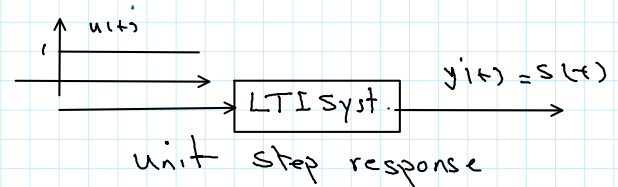
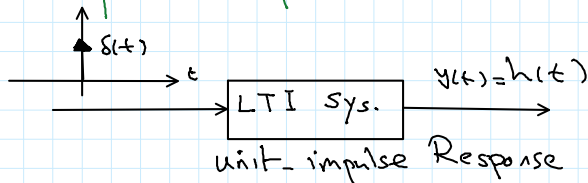
$$II) \quad y(t) = \int_{t-T_1}^{t+T_2} x(\tau) d\tau \quad \text{LTI} \quad h(t) = u(t+T_2) - u(t-T_1)$$

For $T_1 > 0$ & $T_2 > 0$

$$h(t) = \int_{t-T_1}^{t+T_2} \delta(\tau) d\tau = \begin{cases} 1 & \text{if } t+T_2 > 0 \text{ \& } t-T_1 < 0 \\ 0 & \text{o/w} \end{cases}$$



Impulse Response



$$i(t) = i_R(t) + i_C(t)$$

$$i_C(t) = C \frac{dv_C(t)}{dt} = C \frac{dy(t)}{dt}$$

$$i_R(t) = \frac{1}{R} v_C(t) = \frac{1}{R} y(t)$$

II

Replace II into I:

$$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{C} x(t)$$

input-output differential Eq.

Find Impulse Response:

Let initial condition $y(0^-) = 0$

$$y(t) = \int_{0^-}^t \frac{1}{C} e^{-\frac{1}{RC}(t-\lambda)} x(\lambda) d\lambda \quad \text{for } t \geq 0$$

$x(\lambda) = \delta(\lambda)$ & use sifting property

$$y(t) = h(t) = \frac{1}{C} e^{-\left(\frac{1}{RC}\right)t} u(t)$$

$s(t)$ unit step response

$$s(t) = h(t) * u(t)$$

$$\begin{aligned} \frac{ds(t)}{dt} &= \frac{d}{dt} h(t) * u(t) = h(t) * \frac{d}{dt} (u(t)) \\ &= \dot{h}(t) * u(t) = \underline{h(t) * \delta(t) = h(t)} \end{aligned}$$

$\therefore$ impulse Response $\equiv h(t) = \frac{d}{dt} \{ \text{step response} \}$

Also: $s(t) = \int_0^t h(\lambda) d\lambda$

Block Diagram representation:

(ex) $y[n] + \alpha y[n-1] = b_0 x[n] + b_1 x[n-1]$

$$y[n] = -\alpha y[n-1] + b_0 x[n] + b_1 x[n-1]$$

